

INSTALLATION PROCEDURES, EQUIPMENT LIMITATIONS AND OPERATING INSTRUCTIONS

FOR

**ELECTRONIC FLIGHT INSTRUMENT
ASTRONAUTICS P/N 195000-11
CONTROL PANEL
ASTRONAUTICS P/N 195010-11**

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	Number of Sheets 28

Revision History

Rev	Description	Date	Author
A	Initial Release	2007-05-04	J.Cabell

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List of Acronyms

AC	Aircraft Configuration or Alternating Current
ADF	Automatic Direction Finder
ADI	Attitude Director Indicator
AMLCD	Active Matrix Liquid Crystal Display
ARINC	Aeronautical Radio Incorporated
ATP	Acceptance Test Procedure
ATT	Attitude
ATR	Acceptance Test Report
BIT	Built In Test
BI/VI	Bus Interface/Video Interface
BRT	Brightness
C	Celsius
CADI	Copilot Attitude Director Indicator
CCA	Circuit Card Assembly
CHK	Check
CHSI	Copilot Horizontal Situation Indicator
CIWC	Central Instrument Warning Computer
CP	Control Panel
CPU	Central Processor Unit
CRC	Cyclic Redundancy Check
CTR	Center
CWX	Color Weather Radar
DC	Direct Current
DH	Decision Height
DME	Distance Measuring Equipment
EFI	Electronic Flight Instrument
EFIS	Electronic Flight Instrument System
EGPWS	Enhanced Ground Proximity Warning System
ESD	Electrostatic Discharge
EMI	Electromagnetic Interference
ETA	Estimated Time of Arrival
FAA	Federal Aviation Administration
fc	Foot-Candle
fL	Foot-Lambert
FLS	Field-Loadable Software
FD	Flight Director
FMS	Flight Management System
FPGA	Field Programmable Gate Array
FPU	Floating Point Unit
GND	Ground
GPS	Global Positioning System
HSI	Horizontal Situation Indicator
Hz	Hertz
ICD	Interface Control Document
IO	Input/Output
ILS	Instrument Landing System
INS	Inertial Navigation System
IRU	Inertial Reference Unit
LCD	Liquid Crystal Display
MAP	Map Overlay
mW	Milliwatts
N/A	Not Applicable
NAV	Navigation
NM	Nautical Mile

PADI	Pilot Attitude Director Indicator
PHSI	Pilot Horizontal Situation Indicator
PMS	Performance Management System
P/N	Part Number
RALT	Radio Altimeter
RAM	Random Access Memory
RGB	Red Green Blue
RTE	Route
RTN	Return
SRAM	Static RAM
SRD	Software Requirements Data
SRU	Shop Replaceable Unit
S/W	Software
TCAS	Traffic Collision and Avoidance System
TERR	Terrain
TFC	Traffic
TSO	Technical Standard Order
TSOA	Technical Standard Order Authorization
V	Volt
VER	Version
VIOC	Video Input/Output Controller
VOR	Very High Frequency Omni-range Station
VRAM	Video RAM
W	Watts
WXR	Weather Radar
WYPT	Waypoint
XLINK	Crosslink

1 Scope

The following document contains Operating Instructions, Equipment Limitations and Installation Procedures and a description of an Astronautics Electronic Flight Instrument (EFI) and Control Panel (CP). The document has been prepared by the Astronautics Corporation of America (Astronautics) as a supporting document in its request for Technical Standard Order (TSO) Authorization approval from the Federal Aviation Administration (FAA).

This document applies to the Astronautics Electronic Flight Instrument and Control Panel, consisting of the following:

Electronic Flight Instrument	Astronautics' Part Number 195000-11
Control Panel	Astronautics' Part Number 195010-11

Note

Astronautics' part number 195010-11 CP is a "minor change" to Astronautics' 195010-3 CP that has been granted a Technical Standard Order Authorization (TSOA) approval. (Refer to the Title 14 of the Code of Federal Regulations, Part 21, Subpart O, paragraph 21.611 "Design changes").

1.1 EFI and CP Requirements

Detailed software design requirements for the Electronic Flight Instrument and Control Panel are contained in the Software Requirements Data (SRD) document 195544-()\DOC\SRD\SRD.DOC. The () refers to four digits corresponding to the software version optionally followed by a letter revision. If there is a letter revision, it indicates a document change only. The Interface Control Document, ICD10241, also contains design requirements.

1.2 Compliance and Equipment Limitations

The EFI has been tested to environmental and EMI categories of DO-160D as specified in the EFIS DO-160D Environmental Qualification Form.

The CP, on which this design was based, has been tested to environmental and EMI categories of DO-160D as specified in the EFIS DO-160D Environmental Qualification Form and the resulting control string is as follows and will appear on the instrument identification plates.

DO-160D Env. Cat. [(A1)(B1)]-CAB[HB2R]XXXXXXZAAAZYM[A3Z33]XXA

The flight mode embedded software for the EFI on which this design was based has been tested and documented to DO-178B Level A.

The CP contains only discrete circuitry and does not contain any software.

The conditions and tests required for TSO approval of this article are minimum performance standards. It is the responsibility of those installing this article either on or within a specific type or class of aircraft to determine that the aircraft installation conditions are within the TSO standards. The article may be installed only if the installation is performed in accordance with the applicable airworthiness and production requirements.

The EFI and/or CP may be installed only if the installation is performed in accordance with the applicable airworthiness and production requirements.

1.2.1 Field Loadable Software

The EFIS has been certified for field loadable software in accordance with Order 8110.49, Chapter 5 Approval of Field-Loadable Software (FLS). Once approved by the FAA, notice of a software update will be issued in the form of a Service Bulletin. The Service Bulletin will describe the software update and provide instructions for accomplishing the update.

Field-installed software updates are verified for proper installation when complete. Software version and Cyclic Redundancy Check (CRC) numbers are verified in the EFI Maintenance Mode against version and CRC numbers listed in the applicable Service Bulletin.

1.2.2 Instructions for Continued Airworthiness

NOTE TO MAINTENANCE

Include the following instructions in the Aircraft Maintenance Manual or the Instructions for Continued Airworthiness:

“When installing software updates proceed in accordance with the provided Service Bulletin for verification of the update. When performing other general equipment maintenance, verify EFI configuration and validity in accordance with section 5.3.”

WARNING TO MAINTENANCE

Service personnel are to obey standard safety precautions to prevent personal injury while installing or doing maintenance on this unit.

1.2.2.1 Maintenance and Calibration

The EFI and CP contain no items that require specific periodic maintenance or calibration. The only item with a defined limited life is the fluorescent backlight used for illumination of the Active Matrix Liquid Crystal Display (AMLCD) found on the EFI. The typical life of this item is approximately 10,000 powered hours to a level half of its original brightness. Active compensation within the EFI will typically extend the useful life well in excess of that time. Actual life, however, is influenced by a number of factors including, but not limited to power cycles, average operating brightness and temperature. Diminished readability of the display in typical daylight ambient lighting is a likely indication that the display needs servicing to replace the backlight.

NOTE TO MAINTENANCE

Cleaning of the EFI glass may be done as required. Use only a soft lint free cloth with an ordinary glass cleaner to wipe the surface. Do not use soap and water mixtures or brushes to clean the displays. Do not pour or spray cleaning solutions directly on the equipment. Apply the glass cleaner to the cloth then proceed to gently clean the display.

1.2.2.2 Troubleshooting

Table 1.21 describes some potential problems and possible causes along with steps to be taken for troubleshooting and remedial action. To remove the EFI or CP, see the Removal Procedures of paragraphs 4.7 and 4.8.

Table 1.21 Troubleshooting

Problem	Possible Cause	Remedial Action
Blank (Black) Display	No power to EFI	Verify breaker power and that power is applied to appropriate mating pins. Refer to the ICD for the appropriate power pins.
	EFI Failure	Remove instrument for repair.
Blank (White) Display	Internal EFI power supply shutdown	Cycle power to unit; remove unit for repair if problem persists.
‘CHK EFI’ message on EFI	Internal EFI BIT Failure	Run IBIT test in Maintenance Mode as detailed in Paragraph 6.4.4; Remove unit for repair.
Maintenance Page Only Displayed (Instead of normal flight mode)	Invalid EFI configuration strapping	Compare CONFIG code displayed on unit to intended installation configuration (Reference Paragraph 5.3). Remove unit for repair if cabling verified correct

Problem	Possible Cause	Remedial Action
Missing or failed display information	Incorrect Control Panel Settings	Verify Control Panel selections
	Lack or failure of source information	Verify presence of source signal(s) including required validity or conditional signals (Confirm on alternate location EFI and/or replacing unit)
	Internal EFI Failure	Run IBIT test in Maintenance Mode as detailed in Paragraph 6.4.4

2 Applicable Documents

2.1 Astronautics Documents

Table 2.11 Astronautics Documents

Document Number	Document Title
195001-11	EFI Installation Drawing
195011-11	Control Panel Installation Drawing
PLG09400	Pilot's Guide for the EFIS
SRD195544	Software Requirements Data
ICD10241	Interface Control Document for the EFI

Note: Contact Astronautics for the latest revision of the documents listed above

2.2 Industry Standard Documents

DO-160D	Environmental Conditions and Test Procedures for Airborne Equipment
DO-178B	Software Considerations in Airborne Systems and Equipment Certifications
ARINC: 407	Synchro Systems Manual
ARINC: 408	Air Transport Indicator Cases and Mounting
ARINC: 429	Mark 33 Digital Information Transfer System
ARINC: 453	Airborne Weather Radar Digital Data
ARINC: 561	Air Transport Inertial Navigation System
ARINC: 568	Mark 3 Airborne Distance Measuring Equipment
ARINC: 708	Airborne Weather Radar
ARINC: 735	Traffic Alert and Collision Avoidance System
TSO-C113	Technical Standard Order, Airborne Multipurpose Electronic Displays, AND, under the umbrella of TSO-C113 (for display elements only):
TSO-C4c	Bank and Pitch Instruments (indicating gyro-stabilized)
TSO-C6d	Direction Instrument, Magnetic (gyro stabilized type)
TSO-C34e	ILS Glide Slope
TSO-36e	Airborne ILS Localizer
TSO-C40c	VOR Receiving Equipment Operating Within The Radio Frequency Range Of 108-117.95 Megahertz (Mhz).
TSO-C52b	Flight Director Equipment
TSO-C63c	Airborne Weather Radar Equipment
TSO-C92c	Airborne Ground Proximity Warning Equipment
TSO-C119b	Traffic Alert and Collision Avoidance System (TCAS), Airborne Equipment, TCAS II
TSO-C129a	GPS Equipment

3 EFI Description and Operation

3.1 System Architecture

The EFIS consists of four identical EFIs and two CPs. One EFI in Attitude Director Indicator (ADI) mode, one EFI in Horizontal Situation Indicator (HSI) mode and one Control Panel will be located on each side of the cockpit. Many of the aircraft systems are redundant as number one and number two systems. The pilot's EFI and CP typically interface directly with the number one aircraft systems and the copilot's EFI and CP interface directly with the number two aircraft systems. Systems that are a single source typically interface to both sides of the cockpit.

The EFIs are 5ATI units, which interface with numerous aircraft systems utilizing ARINC 429, ARINC 453 and ARINC 568, discrete, analog, and RS422 signals. The CP contains discrete rotary knobs and pushbuttons that connect directly to the EFI for user interface.

The EFI uses a 256 shade of gray 4"x4" AMLCD of which 64 shades are used for the display presentation. The EFI bezel has two pushbuttons for brightness control and an inclinometer (see Figure 3.31). The CP has nine (9) pushbuttons and two (2) rotary knobs for control of EFI functionality such as Range and Mode selections (see Figure 3.32).

Table 3.11 EFI and CP Specifications

	EFI	CP
Operating Voltage	28VDC	-
Lighting Voltage	5VAC	5VAC
Ambient Operating Temperature	-20° to + 55°C	-20° to + 55°C
Overall Dimensions (inches)		
Length	9.65	3.90
Height	5.10	2.25
Width	5.10	5.75
Dimensions Excluding Bezel and Connectors (inches)		
Length	8.50	3.00
Height	4.98	2.15
Width	4.98	4.91
Weight lbs. – maximum	8.5	2
Connector Contacts Keyed		
J1 55 N	MS27508E22F55P	N/A
J2 79 N	MS27508E20A35P	N/A
J3 79 A	MS27508E20A35PA	N/A
J1 41 N	N/A	MS27508E20F41P

3.2 EFI and CP Design

3.2.1 EFI Design and Construction

The EFI is constructed of two major assembly components – a Display Assembly and a Chassis and Interconnect Assembly. When assembled, the EFI meets the physical envelope specified in ARINC408A for a 5ATI case. For complete dimensions reference the EFI Installation Drawing 195001-11.

3.2.2 Display Assembly

The Display Assembly is based on a 4"x4" usable area color AMLCD. The display is illuminated with a serpentine hot cathode fluorescent lamp that uses a diffuser and optical films to achieve uniform illumination. The lamp is controlled and driven by the Inverter Interface and Inverter Circuit Card Assemblies (CCAs). The operator adjusts

the lamp illumination with two momentary pushbuttons on the front bezel to increase (BRT) and decrease (DIM) inputs to the lamp Inverter circuitry.

3.2.3 Display Resolution

The AMLCD provides a resolution of 524x524 color pixels. The EFI design presents a 4"x4" display area with a minimum usable resolution of 512x512 pixels.

3.2.4 Pixel Configuration

The pixel configuration is a color group configured in a red, green, and blue vertical stripe pattern. The red, green and blue sub-pixels are individually addressable.

3.2.5 Luminance

EFI display luminance is adjustable from the front bezel pushbuttons. The range of adjustment is from black (<0.1 fL) to a minimum of 120 fL white.

3.2.6 Color Palette

The EFI is capable of displaying each of the red, green, and blue (RGB) primary colors in 64 shades. In addition, the EFI is capable of displaying 262,144 mixed colors, which result from all the possible combinations of the shades available for each primary color. The display background (off pixels) shall be black.

3.2.7 Primary Color and Reference White Chromaticities

The area-averaged chromaticities of the RGB primary color sub-pixels fall on or within the R, G, or B chromaticity coordinates listed below.

Table 3.21 Chromaticity

COLOR	u'	v'
Red	0.378	0.504
Green	0.137	0.545
Blue	0.148	0.280

3.2.8 Contrast

Table 3.22 defines the EFI viewing angles and contrast ratios in a room ambient.

Table 3.22 Viewing Angle and Contrast Ratio

Horizontal Angle	Vertical Angle	Contrast
0	0	>40:1
0	+30	>30:1
0	-5	>30:1
+/- 30	0	>20:1
+/- 45	0	>10:1

When exposed to ambient illumination conditions of up to 8,000 fc, the display primary color chromaticities are clearly identifiable as red, green, and blue within the viewing cone and at the design eye position.

3.2.9 Display Defects

A sub-pixel shall be considered defective when it is failed permanently or sporadically in either the "bright" emitting state or the "dark" non-emitting state. The sum of "bright" and "dark" defective sub-pixels shall not exceed ten (10) within the viewing area of the AMLCD. There shall be no full rows or columns of defective sub-pixels. A group of two defective adjacent color sub-pixels shall be considered a cluster defect. There shall be no more than one (1) cluster defect within the viewing area of the display. The EFI is inspected to ensure that existing defects will not affect legibility of critical data. There shall be no bubbles or scratches visible to the unaided eye.

3.2.10 Display Warm-Up Time

The EFI contains an AMLCD heater and backlight lamp heater with associated control circuitry to reduce the time to a usable display from -20°C startup. Typical time to full brightness of an actively moving display is less than 10 minutes.

Table 3.23 EFI Power

EFI pin	Signal	Description	Power
J1 - C, D, Y	28VDC	Primary Power	90W cold start 45W nominal
J1 - E, F, SB	28VDC RTN		
J1 - K	5VAC	Lighting Power	500mW
J1 - L	5VAC RTN		

The maximum power occurs at cold temperature turn-on when the AMLCD and lamp heaters are operating. These will shut off when the internal display temperature reaches approximately 10°C.

The EFI contains holdup circuitry to maintain EFI operation through a single 50-millisecond voltage dropout. Consecutive dropouts require a recharge time for the internal energy storage.

3.2.11 CP Lighting Power

The Control Panels require 5VAC lighting power to illuminate the switch panel legends. Lighting power requirement is defined below:

Table 3.24 CP Power

CP Pin	Signal	Description	Power
J1 - J, c	5VAC	Lighting Power	7.5W
J1 - K, q	5VAC RTN		

3.2.12 EFI Input/Output

The EFI is designed with multiple discrete, analog and digital input interfaces. They are designed to interface with a variety of standard aircraft system signals such as ARINC407-1, ARINC568, ARINC429, ARINC453 and RS422 as well as analogs and discretes. Also, transient suppressors and filters are designed into the EFI to withstand the test categories listed in Table 3.41.

Table 3.25 EFI Interfaces

Type of IO		Total Capability
Synchro Inputs		4
Analog Inputs		10
Discrete Inputs	28V/Open	23
	Gnd/Open	48
Analog Outputs (400Hz AC)		2
Discrete Outputs	28V/Open	4
	Gnd/Open	2
ARINC 429 Input		8
ARINC 429 Output		2
ARINC 453 Input		2

Type of IO	Total Capability
RS-422 Receive	5
RS-422 Transmit	2
ARINC 568 Input	1

3.2.13 EFI and CP Connectors

The design and construction of the EFI and CP incorporates features such that they are mechanically impossible to install incorrectly. These features apply to attachment cables, lamp connectors, electrical plugs and any such items associated with the installation in the host display unit. Features such as mechanically keyed mating, different size connectors, and/or clearly labeled connectors are incorporated to eliminate any possibility of incorrect installation.

The mechanical interfaces to the connectors are as follows:

EFI J1 - MS27508E22F55P, mates with MS27473E22F55S or equivalent.

EFI J2 - MS27508E20A35P, mates with MS27473E20F35S or equivalent.

EFI J3 - MS27508E20A35PA, mates with MS27473E20F35SA or equivalent.

CP J1 - MS27508E20F41P, mates with MS27473E20F41S or equivalent.

The display portion of the EFI is based on a 4"x4" usable area active matrix color display. The display is illuminated with a serpentine hot cathode fluorescent lamp that uses a diffuser and optical films to achieve uniform illumination. The operator adjusts the lamp illumination with two momentary pushbuttons on the front bezel to change the input to the lamp inverter circuitry. Figure 3.21 provides a block diagram of the typical Astronautics EFI architecture.

The backend electronic portion of the EFI contains a set of Shop Replaceable Units (SRUs), which provide the following functions:

Connector CCA:	contains the external aircraft connectors and circuitry for signal filtering and lightning protection
Input/Output #1 CCA:	provides analog, synchro and discrete input and output circuitry
Input/Output #2 CCA:	provides analog, synchro and discrete input and output circuitry
Serial Interface CCA:	provides the input and output circuitry for the ARINC 429, ARINC 568 and RS422 serial interfaces
Color Weather Radar CCA:	contains input circuitry to process ARINC 453 interfaces
Low Voltage Power Supply CCAs:	a two card set which takes the 28VDC aircraft power and generates the required internal system voltages
Inverter CCA:	controls voltages for the lamp and heater
Backplane CCA:	the system interconnect board which the other SRU's interface with.
Graphics Engine and CPU CCAs:	contain a Pentium processor, program memories and display memories, which utilize an embedded program to perform the IO processing and graphics generation for the EFI

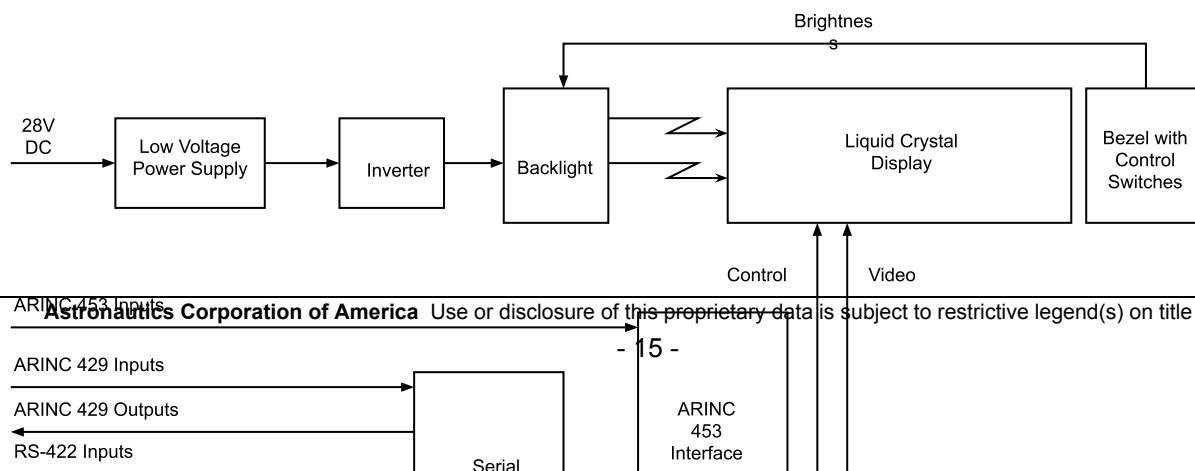


Figure 3.21 EFI Block Diagram

3.3 Operating Instructions

3.3.1 Electronic Flight Instrument

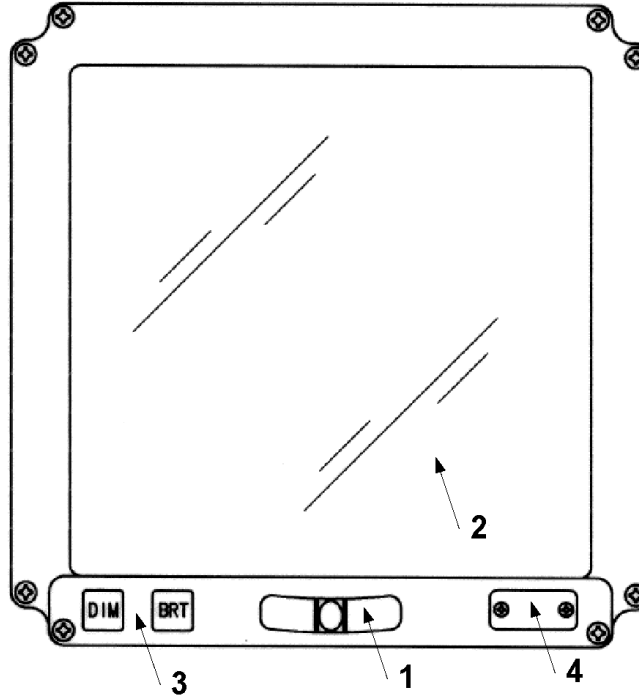


Figure 3.31 Electronic Flight Instrument

Table 3.31 EFI Features

Index	Name	Function
1	Inclinometer	Mechanical Slip Indicator
2	Viewing Screen	A 4x4 inch Active Matrix Liquid Crystal Display
3	Brightness Controls	Pressing the BRT pushbutton increases the brightness and pressing DIM decreases the brightness. The display cannot be dimmed to a completely black screen (OFF).
4	Maintenance Port Cover	Removing the two screws and plate will reveal the 8-pin Maintenance Port. The Maintenance Port can be used to load software in the field or view diagnostic data.

3.3.2 Control Panel

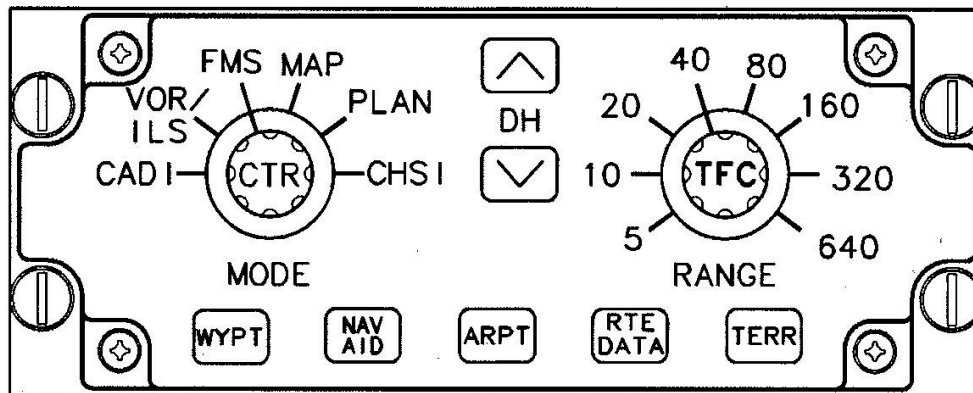


Figure 3.32 Control Panel

3.3.2.1 Mode Control Knob

Pressing the inner pushbutton toggles the HSI compass between Center (360° compass centered on the aircraft) and Arc (120° compass with the aircraft at the bottom). This is only used in FMS, MAP and VOR positions.

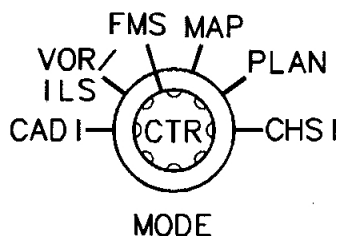


Figure 3.33 Mode Knob

3.3.2.1.1 VOR/ILS Mode

The VOR/ILS mode selects VOR or ILS as the source of navigation with the heading referenced to magnetic north for display on the EHSI. VOR/ILS navigation is available in either ARC or CTR mode. The VOR/ILS position display is heading up and displays track, heading, true airspeed, ground speed, and wind from the FMS with VOR or ILS navigation information. TERR and TCAS may also be displayed.

3.3.2.1.2 FMS Mode (FMS Navigation)

The FMS display mode is available in either ARC or CTR mode. The FMS display mode is heading up and displays track, heading, ground speed, true airspeed, ETA, and wind with FMS navigation information. TERR and TCAS may also be displayed.

3.3.2.1.3 MAP Mode (FMS Navigation)

The MAP display mode is available in ARC or CTR mode. When in MAP Mode, the FMS furnishes the navigation information. The MAP display mode is track up and presents information against a moving map background and includes track, heading, route, wind, distance, ETA, true airspeed, ground speed, position trend, and selected navigation data points. TERR and TCAS may also be displayed.

3.3.2.1.4 PLAN Mode (FMS Navigation)

Plan Mode presents a planning mode with North up in a fixed position. Heading will not be displayed. Planned course information will be displayed. When PLAN is selected on the Control Panel, the EADI will switch automatically to the Combined Mode.

3.3.2.1.5 CADI Mode

The CADI position enables the selection of a combined ADI and HSI on the EADI with VOR/ILS as the navigation source and magnetic heading reference.

3.3.2.1.6 CHSI Mode

The CHSI position enables the selection of a combined ADI and HSI on the EHSI with VOR/ILS as the navigation source and magnetic heading reference. The differences from the CADI position are that the offside DH alert is not present and the digital readout of the onside Decision Height is displayed only as three dashes.

3.3.2.2 Range Knob

Turning the outer knob selects 5, 10, 20, 40, 80, 160, 320 or 640NM for the maximum displayed range on the HSI (i.e. map or terrain mode). Range is selected for VOR/ILS, FMS, MAP, and PLAN modes.

Pressing the TFC pushbutton alternately displays and removes the traffic overlay on the EHSI.

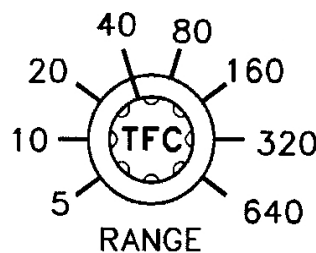


Figure 3.34 Range Knob

3.3.2.3 Terrain Pushbutton

The TERR pushbutton is used to select/de-select terrain monitoring from the EGPWS



Figure 3.35 Terrain Pushbutton

3.3.2.4 Map De-Clutter Pushbuttons

Pressing these pushbuttons alternately displays and removes supplementary symbols from the map overlay.

- WYPT used to select/de-select off-route waypoint data.
- NAV AID used to select/de-select VORs, and other navigation aids as furnished by the FMS.
- ARPT used to select/de-select display of near airports as furnished by the FMS.
- RTE DATA used to select/de-select en route data from the FMS.



Figure 3.36 Map De-Clutter Pushbuttons

3.3.2.5 Decision Height Pushbuttons

Two DH select switches are provided. They are used to select the Decision Height for approaches. The upper switch increases the displayed Decision Height and the lower switch decreases the displayed Decision Height. The minimum selectable Decision Height is 0 feet and the maximum selectable Decision Height is 500 feet.

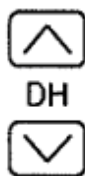


Figure 3.37 Decision Height Pushbuttons

3.4 Environmental, Power, EMI and Lightning Compliance Categories

The EFI and CP on which this design was based were tested and qualified to the following DO-160D categories:

Table 3.41 Qualification Testing

TSO-C113 Test Description	DO-160D Section and Category
Temperature Altitude	Section 4, Categories A1 (controlled temperature pressurized to 15,000 foot altitude or less) and B1 (controlled temperature but NOT pressurized, up to 25,000 foot altitude) plus altitude to 45,000 ft.
Low Temperature	Tested to –55 deg C (ground survival) and –20 deg C (operational)
High Temperature	Tested to +85 deg C (ground survival) and +70 deg C (short-time operational) Tested to +55 deg C (operational)
Altitude Test	Tested to 45,000 feet
Decompression Test	Tested to 8,000 feet and changing to 45,000 feet within 15 seconds
Overpressure Test	Tested to –15,000 feet for 10 minutes
Temperature Variation Test	Section 5, Category C
Humidity Test	Section 6, Category A
Shock Tests	Section 7, Paragraph 7, tested to Category B requirements.
Operational Shock, Impulse	Paragraph 7.2.1: eleven millisecond, six g terminal saw-tooth shock pulses.
Crash Safety, Impulse	Paragraph 7.3.1: eleven millisecond, 20 g terminal saw-tooth shock pulses.
Crash Shock, Sustained	Paragraph 7.3.2: Category B for fixed-wing aircraft, in a “known and fixed” orientation.
Vibration Tests	Section 8, HB2R
Explosion Test	Section 9, Category X, no test required
Water Proofness Test	Section 10, Category X, no test required.
Fluid Susceptibility Test	Section 11, Category X, no test required.
Sands and Dust Test	Section 12, Category X, no test required.
Fungus Resistance Test	Section 13, Category X, no test required.
Salt Spray Test	Section 14, Category X, no test required.

TSO-C113 Test Description	DO-160D Section and Category
Magnetic Effect Test	Section 15, Category Z.
Power Input Test: Normal & Abnormal	Section 16, Category A for DC voltage equipment.
Voltage Spike Conducted Test	Section 17, Category A.
Audio Frequency Conducted Susceptibility Test	Section 18, Category A for DC voltage equipment.
Induced Signal Susceptibility Test	Section 19, Category Z.
Radio Frequency Susceptibility Test	Section 20, Category Y.
Emission of Radio Frequency Energy Test.	Section 21, Category M for both Radiated and Conducted Emissions.
Indirect Lightning	Section 22, Category A3Z33 J33 – EFI J1, J2, J3 G33 – CP J1
Lightning Direct Effects	Section 23, Category X.
Icing	Section 24, Category X.
Electrostatic Discharge (ESD)	Section 25, Category A.
X-ray Radiation	Not a DO-160 test (required by TSO-C113). Analysis per AS 8034 para. 5.20. (The AMLCD does not produce X-ray radiation).
U.V. Radiation	Not a DO-160 test (required by TSO-C113). Tested to AS 8034 para. 5.21.
Fogging	Not a DO-160 test (required by TSO-C113). Not required per AS 8034 para. 5.22 (not hermetically sealed).
Thermal Shock	Not a DO-160 test (required by TSO-C113). Tested to AS 8034 para. 5.23.
Dielectric Test	Not a DO-160 test (required by TSO-C113). Tested to AS 8034 para. 5.24.

3.5 Environmental Stress Screening

Each EFI and Control Panel shall undergo Environmental Stress Screening, which includes vibration and temperature cycles prior to the final acceptance test and shipment.

3.6 Acceptance Test

Prior to delivery each EFI and Control Panel shall have successfully passed an Acceptance Test Procedure (ATP) that verifies the functionality of the units in a bench ambient. An Acceptance Test Report (ATR) is provided with each unit shipped.

4 Installation Procedures

4.1 Package Contents

The EFIs and Control Panels are packaged and shipped in cartons and may be packaged separately. A packing slip will be included which identifies the contents of each parcel.

Within the carton is the deliverable unit and documents. There are no fasteners, connectors, cables or other materials supplied with either unit.

The EFIs and Control Panels are each shipped in a plastic bag with protective caps on the connectors. The EFI also has a protective cover on the front of the AMLCD.

Each EFI and Control Panel is shipped from the manufacturer with, as a minimum, an ATR, which is the result of the final test on the unit.

4.2 Unpacking the Equipment

CAUTION TO MAINTENANCE

This unit contains electrostatic discharge (ESD) sensitive components and assemblies that can be damaged by static voltages. Although most components and assemblies contain internal protection circuits, proper ESD handling procedures should always be practiced.

4.2.1 Unpacking the Electronic Flight Instrument

The plastic bag, connector caps and AMLCD protective cover should be left in place until the EFI is ready to either bench test or install in the aircraft. The front surface of the EFI display is a heater glass. While the glass provides protection for the AMLCD, it should be handled with appropriate care in order to avoid pressure on the surface. Sharp tools should never be used near the front surface of the display.

4.2.2 Unpacking the Control Panel

The plastic bag and connector caps should be left in place until the Control Panel is ready to either bench test or install in the aircraft.

4.3 Installation Considerations

As with all electronic equipment, prior to installation the connector protective caps should be removed and the connectors inspected for bent pins. If the AMLCD protective cover is still in place, it should be left on until the unit is installed. Care should always be taken to minimize contact with the display surface.

CAUTION TO MAINTENANCE

Excessive pressure on the AMLCD surface can cause damage.

See section 1.2.2.1 Maintenance and Calibration for cleaning procedures and precautions.

The knobs on the CP are ruggedized, but as with all controls care should be taken to not lift or pull on these as the shafts may break.

WARNING TO MAINTENANCE

Power must be present only on the connector pins specified in the Interface Control Document. Extensive damage can result to the unit or maintenance personnel if power is present on any other pins.

4.3.1 Air Circulation

The EFI is cooled both by convection and radiation. Maximum case temperature for long-term reliable operation should be limited to 60°C to the extent practical. Maximum case temperature for safe long-term operation should be limited to 77°C; however reliability will be degraded if run at this temperature on a continuous basis. Maximum case temperature for short duration (less than 30 minutes) should under no circumstances exceed 85°C. The maximum case temperature of an EFI in an open ambient occurs on the top-center of the case approximately one inch back from the front mounting surface. Temperature variation between the front and rear of the case is typically 5°C.

There are no cooling considerations for the identified Control Panel. The only dissipated power is in the lighting panel, which is cooled by convection and radiation within the cockpit environment.

4.3.2 Weight and Balance

The size and balance of the EFI are shown in the Installation Drawing P/N 195001-11. The weight is defined in Table 3.11 of this document.

The size and balance of the Control Panel are shown in the Installation Drawing P/N 195011-11. The weight is defined in Table 3.11 of this document.

4.3.3 Cabling and Wiring

The EFI completed its TSO EMI testing with a specific cabling configuration. The CP, on which this design was based, has completed its TSO EMI testing with a specific cabling configuration. The cabling was constructed to represent, as close as practical, the intended installation for wire type and shielding and should be considered in the actual aircraft installation. Shield termination should be in accordance with applicable aircraft standards. Consult document number ICD10241 for a complete listing of signal descriptions, wire recommendations and pinout of the EFI and CP.

4.4 Installation Procedure for the Electronic Flight Instrument

1. Ensure all electrical power to EFI is off.
2. Remove and retain plastic end caps.
3. Connect aircraft cable to connector J1 on rear of EFI. Connector J1 mates with connector MS27473E22F55S or equivalent.
4. Connect aircraft cable to connector J2 on rear of EFI. Connector J2 mates with connector MS27473E20F35S or equivalent.
5. Connect aircraft cable to connector J3 on rear of EFI. Connector J3 mates with connector MS27473E20F35SA or equivalent.
6. Carefully slide EFI with basket clamp into mounting hole.
7. Alternately tighten each screw to secure EFI to mounting surface.
8. Repeat the above steps for the remaining Electronic Flight Instruments.
9. Apply power to the EFIs and perform power up check in accordance with section 4.6.

4.5 Installation Procedure for the Control Panel

1. Ensure all electrical power to CP is off.
2. Connect aircraft cable to connector J1 on rear of CP. Connector J1 mates with connector MS27473E20F41S or equivalent.
3. Carefully slide CP into mounting hole, and then alternately tighten each captive screw to secure CP to mounting surface.
4. Repeat the above steps for the remaining Control Panel.
5. Apply power to the CPs and perform power up check in accordance with section 4.6.

4.6 Electronic Flight Instrument Checkout

The installation checkout is to be performed on the ground.

NOTE

The maintenance port behind the access panel located on the front of the EFI is not a part of the normal aircraft checkout procedure. The port can be used with assistance from Astronautics to provide additional aircraft diagnostics in special cases or to load software upgrades in the field. See Figure 3.31 for location of the maintenance port.

1. Apply aircraft power to the EFIs and CPs.
2. Verify the EFI powers up with a valid HSI type display.
3. Operate CP controls and verify operation as described in section 3.3.2.
4. Operate the BRT and DIM pushbuttons on the EFI and note the display brightness change.
5. Vary the cockpit lighting bus (located on the cockpit instrument panel) and note the bezel lighting on the EFI and CP increase and decrease.
6. With the full EFIS installed, verify the following flags are not present (CHK EFI; XLINK FAIL)

Note

The EFI tracks certain sensor failures by removing the affected display elements. To be able to evaluate these elements, it is necessary to supply valid sensor inputs.

Note

If there is no Control Panel installed, the EFI will revert to its default settings.

4.7 Electronic Flight Instrument Removal

WARNING TO MAINTENANCE

Disconnect all power to unit before removing it. Removing the unit with power connected is dangerous to life and may cause voltage transients that can damage the unit.

This procedure is to be conducted by maintenance personnel on the ground.

1. Ensure the power is removed from the aircraft cables prior to disconnecting the EFI.
2. Loosen the connections to the basket clamp holding the unit to the panel.
3. Slide the EFI forward to remove it from the panel.
4. Holding the EFI free of the aircraft panel, twist and remove the three aircraft interface cables.
5. Install the plastic end caps on the three connectors.

4.8 Control Panel Removal

1. Ensure the power is removed from the aircraft cables prior to disconnecting the Control Panel.
2. Turn each of the retaining fasteners counterclockwise until the fasteners are loose.
3. When all the fasteners are loose, slide the Control Panel forward to remove it from the aircraft panel.
4. Holding the Control Panel free of the aircraft panel, twist and remove the aircraft interface cable.

5 Equipment Limitations

5.1 Equipment Interfaces

The hardware and software of the EFI display has been designed and tested to interface with a number of standard aircraft interfaces. Some of these interfaces have specific characteristics, such as the information format of a particular ARINC429 label that may make them incompatible with similar equipment. Information is also sometimes displayed in different formats to maintain similarity to a display being replaced by the EFI. To accommodate unique requirements with different interface systems in a single display, configuration-strapping discretes, hard wired into the aircraft installation harness, are used to customize the EFI to a particular installation.

For redundancy purposes, certain information is shared, or crosslinked, between EFIs. Table 5.11 lists the information that is crosslinked between EFIs. See Figure 5.11 below for the typical connections involved cross-linking information between Electronic Flight Instruments.

Table 5.11 EFI Crosslink Data

Pitch	Navigation Source	Terminal Mode
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Roll	Heading Source	Bearing
Heading	Attitude Source	DME Distance
Glideslope	Glideslope Qualified	DH Crossed Flag
Localizer		Localizer Capture
Radio Altitude		

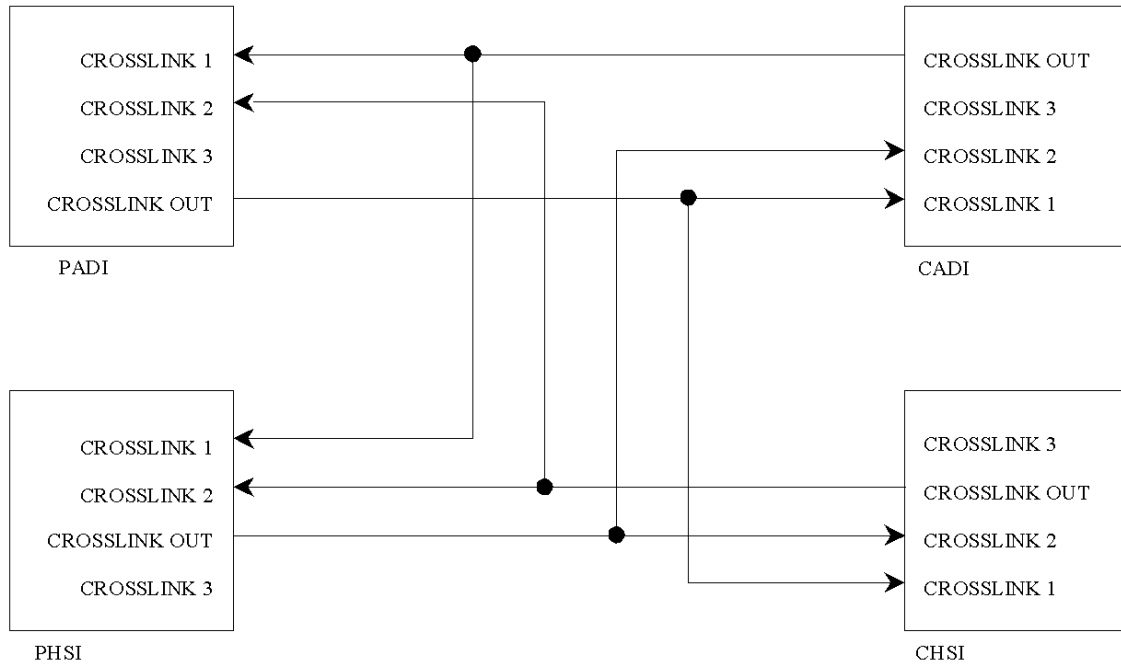


Figure 5.11 EFI Cross-link Connections

5.2 Interface Capability

The EFI is designed to work with the following generic interfaces and/or equipment. This set of interfacing equipment or equivalent can be considered a limitation on the function of the EFI. This limitation extends only to the currently designed hardware and software at the time of TSO submittal and does not necessarily represent a complete list of the equipment capable of interfacing to the display. Table 5.21 is for reference use only. For additional detail regarding specific interfaces, reference ICD10241 or consult Astronautics Corporation of America for additional information.

Table 5.21 Interface Capability

Interface Equipment	Manufacturer	Model	Interface
Control Panel	Astronautics	195010-11	Discrete
EFI	Astronautics	195000-11	ARINC429, ARINC453, ARINC568, RS-422, Synchro, Analog, Discrete
Central Instrument Warning Computer (CIWC)	Collins	54W-3B	Discrete
Compass Coupler	Sperry	2591201-903	ARINC407-1 Synchro, Discrete
Autothrottle Computer	Marconi-Elliott	42-017-01	Analog
Distance Measuring Equipment (DME)	Collins	860E-5	ARINC 568
Flight Director (FD)	Sperry	2593362-934	Analog, Discrete
EGPWS	Honeywell	Enhanced Mark VII	ARINC 453
Weather Radar (WXR)	Honeywell	RDR-4B	ARINC 453

Interface Equipment	Manufacturer	Model	Interface
Radar Altimeter (RALT)	Collins	860F-4	ARINC 552 Analog, Discrete
Flight Management System (FMS)	Canadian Marconi (BAE Systems)	CMA-900	ARINC 429
Inertial Reference Unit (IRU)	Litton	LTN-92	ARINC 429
Performance Management System (PMS)	Ambac	2-00080-100-2	Analog, Discrete
Marker Beacon Receiver	Collins	51Z-4	Analog
Traffic Collision Avoidance System (TCAS)	Collins	TTR920	ARINC 429
VHF Nav Receiver (VOR)	Collins	51RV-5D	ARINC 407-1 Synchro, Analog, Discrete

5.3 Configuration Strapping

The primary operating mode of an EFI is configurable by aircraft strapping (programming) pins. These strapping pins are also used to identify a system interface configuration where it is required for the display to respond to some interfaces in a unique manner.

The system configurations defined for the display at the time of this TSO submittal are provided in Table 5.31. The configuration pin strapping is defined by indicating which pins are left open (OPEN) or grounded (GND). The setting can be read on the Maintenance Mode page (refer to section 6) of the display. Any strapping other than those listed below will result in the display powering up into Maintenance Mode with an invalid strapping configuration identified (shown in red). See the SRD for further details on configuration options.

Please consult Astronautics Corporation of America for additional information on system configuration options.

Table 5.31 Configuration Strapping #11

Signal Name	EFI pin	PADI	PHSI	CADI	CHSI
PILOT/COPILOT	J3-27	OPEN	OPEN	GND	GND
ADI/HSI	J3-28	OPEN	GND	OPEN	GND
AC0	J3-2	GND	GND	GND	GND
AC1	J3-48	GND	GND	GND	GND
AC2	J3-49	OPEN	OPEN	OPEN	OPEN
AC3	J3-50	GND	GND	GND	GND
AC4	J3-29	OPEN	OPEN	OPEN	OPEN
AC5	J3-30	OPEN	OPEN	OPEN	OPEN
AC6	J2-7	OPEN	OPEN	OPEN	OPEN
PARITY	J3-1	GND	OPEN	OPEN	GND

5.4 TSO Certification

The EFI has been granted TSOA approval. The CP that this design is based on has been granted TSOA approval. The EFI and CP nameplates will reflect the DO-160D categories for the TSOA.

Below is a list of the TSOs, which are applicable to this EFI, have been approved for the design that this was based on, and appear on the instrument identification plates.

- TSO-C113 Technical Standard Order, Airborne Multipurpose Electronic Displays, AND, under the umbrella of TSO-C113 (for display elements only):
- TSO-C4c Bank and pitch instruments (indicating gyro-stabilized)
- TSO-C6d Direction instrument, magnetic (gyro stabilized type)

TSO-C34e	ILS Glide Slope
TSO-C36e	Airborne ILS Localizer
TSO-C40c	VOR Receiving Equipment Operating Within The Radio Frequency Range Of 108-117.95 Megahertz (MHz).
TSO-C52b	Flight Director Equipment
TSO-C63c	Airborne weather and ground mapping pulsed radars
TSO-C92c	Airborne ground proximity warning equipment
TSO-C119b	Traffic alert and collision avoidance system (TCAS) airborne equipment, TCAS II
TSO-C129a	Airborne Supplemental Navigation Equipment Using the Global Positioning System (GPS)

6 Maintenance Mode

On the ground, maintenance personnel can use the Maintenance Mode to provide EFI Configuration data and BuiltIn Test (BIT) results. When VOR/ILS mode is selected, access to Maintenance mode can be accomplished by simultaneously pressing and holding the four Control Panel de-clutter pushbuttons NAV, WYPT, ARPT and DATA.

6.1 Maintenance Display Screen

The following is a description of the Maintenance Display Screen fields and content.

The Maintenance Screen consists of three parts:

1. The top of the screen contains a listing of the software configuration data in the EFI.
2. The second part of the screen is a hardware configuration section.
3. The third part of the screen contains a list of the current internal BIT faults detected.

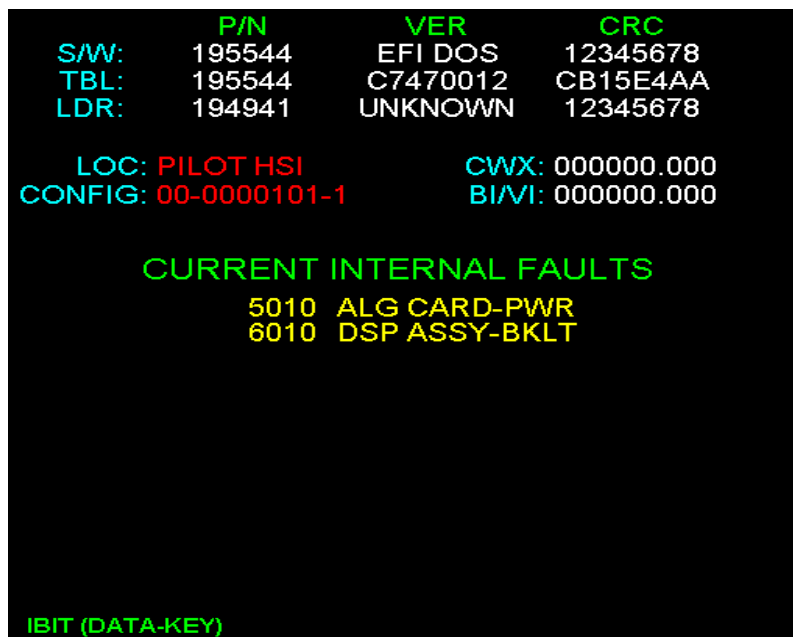


Figure 6.11 Example of Maintenance Display Screen

6.2 Software Configuration

Table 6.21 below contains software configuration data layout. The table has 3 rows and 3 columns of data with an additional column on the left to identify the rows and an additional row at the top to identify the columns. The software configuration information is summarized below:

Table 6.21 Configuration Layout

	Column Label: P/N	Column Label: VER	Column Label: CRC
Row label: S/W	Part Number of EFI software loaded onto the unit. This is a field read from the resident software.	Version of installed flight software.	Code is calculated as part of loading process and compared to stored value. If the CRC does not match the unit continues to try to reload - no screen would be present.
Row label: TBL	Part number of EFI software table (contains fonts, symbol details).	Version of software table read from resident software.	Code is calculated as part of loading process and compared to stored value.
Row label: LDR	Part number of EFI software loader. Read from resident software.	Version of software loader read from resident software.	Code is calculated as part of loading process and compared to stored value.

6.3 System and Firmware Configuration

The middle screen entries contain system and firmware configuration data. The text in the LOC field is developed from strapping, which lists the cockpit location that the display is configured for. CONFIG is a binary listing of the strapping that the EFI is reading. If a configuration problem is detected, the display will automatically power-up into the Maintenance Mode screen. If the strapping configuration is bad, the maintenance operator will not be able to leave this screen until a valid configuration is used (Reference paragraph 5.3 for valid system configuration).

The remaining fields represent the versions of Field Programmable Gate Array (FPGA) code in the unit. The CWX is the color weather FPGA code version. The BI/VI is a part number that relates to both the bus and video interfaces. These versions are read from registers in the FPGAs and are controlled as part of the hardware configuration of the unit. The firmware is maintained only at the depot level.

6.4 BIT Tests and Current Internal Faults

After entering the Maintenance Mode as described in paragraph 6, the initiated built-in test function can be utilized. On the bottom of the screen, IBIT is listed next to the CP pushbutton that will start the test. Pressing the appropriate CP pushbutton will force execution of the IBIT routines and any resulting faults will be indicated in the Current Internal Faults area.

The third part of the screen is the Current Internal Faults list. The first two columns of the data below contain the code and text of faults that could be seen on the Internal Faults portion of the screen.

6.4.1 Built-In Test

The EFI has three levels of Built-In Test: Power-Up, Continuous and Initiated BIT.

6.4.2 Power-Up BIT

Power-Up Bit is performed automatically upon application of primary power to the EFI. This BIT is limited in scope and is completed within a few hundred milliseconds.

6.4.3 Continuous BIT

The EFI performs Continuous BIT. This is a set of BIT tests that are run during normal operation. This includes tests to verify general communication and operation of internal circuit cards as well as tests of certain aircraft interfaces.

6.4.4 Initiated BIT

The EFI performs Initiated BIT when commanded from the Maintenance Mode. This diagnostic BIT is performed during installation or as a maintenance function.

Table 6.41 Fault Codes

SRU	Failed Component	Description	Text	Code	CBIT
CPU	CPU	Verify correct operation of registers, FPU, and cache.	CPU CARD-CPU	0010	X
	Static RAM	Verify correct operation of SRAM.	CPU CARD-RAM	0020	X
	Flash	Verify checksum of program area of flash.	CPU CARD-ROM	0030	X
	Video RAM	Verify correct operation of VRAM.	CPU CARD-VRAM	0050	X
	Flash	Verify checksums of some other areas of flash.	CPU CARD-NVM	0060	
	VIOC	Verify correct operation of buffer swap.	CPU CARD-VIOC	0080	
Analog I/O Card #1 or #2 (x=card number, y=x-1)	FPGA	Verify basic operation of FPGA.	IOx CARD-FPGA	1y10	X
	A-to-D Converter	Verify basic operation of A/D Converter.	IOx CARD-ANLG	1y30	X
	A-to-D Converter	Verify correct operation of A/D Converter by converting a reference value.	IOx CARD-AREF	1y40	X
Serial I/O Card	FPGA	Verify basic operation of FPGA.	SER CARD-FPGA	2010	X
	ARINC-429	Verify correct loopback test of each channel (A-H) that is used in this configuration.	SER CARD-ALA to SER CARD-ALH	2020-2027	
	ARINC-429	Check for absence of sequence errors for each chipset (A-D or E-H) that is used in this configuration.	SER CARD-ASAD to SER CARD-ASEH	2030-2031	X
	ARINC-429	Check for absence of address errors for each chipset (A-D or E-H) that is used in this configuration.	SER CARD-AAAD to SER CARD-AAEH	2040-2041	X
	UART	Verify correct loopback test of each channel (A-H) that is used in this configuration.	SER CARD-ULA to SER CARD-ULH	2050-2057	
CWX I/O Card	FPGA	Verify basic operation of FPGA.	CWX CARD-FPGA	4010	X
Analog Interface Card	LCD Power Supply	Read LCD Power Valid discrete and check if LCD power supply is turned on.	ALG CARD-PWR	5010	X
Display Head Assembly	Backlight	Check for presence of Backlight Valid discrete.	DSP ASSY-BKLT	6010	X